

# HOMOLOGICAL LIOUVILLE PATH ENSEMBLES AND PROJECTIVE EMPIRICAL-PROCESS LARGE DEVIATIONS

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ABSTRACT. Orbit births are handled by a renormalized physical current bundle whose norm scales the vanishing birth mode. A finite family of periodic certificates controls compact frequencies, while a separate geometric nonlocal-integrability theorem supplies Dolgopyat cancellation. The vector-roof local limit theorem is stated in sum variables and has the correct average-window scaling.

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## 1. ROUND-EIGHT RECONSTRUCTION: RENORMALIZED BIRTH BUNDLE, UNI, AND THE CORRECTLY SCALED VECTOR-ROOF LLT

We distinguish three logically different ingredients: the physical quotient through orbit births, compact-frequency aperiodicity, and high-frequency nonlocal integrability. The local limit statement is written in sum variables first and then translated to windows for the normalized roof average.

**1.1. The renormalized physical current bundle.** Let  $E_R : \mathbb{B} \rightarrow \mathcal{D}'$  be the realization of the ambient trace coordinates. At a simple birth  $R_0$  choose

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a smooth ambient splitting  $\mathbb{B} = B^{\text{old}} \oplus B^{\text{b}}$  for which

$$E_R(x, y) = E_R^{\text{old}}x + (R - R_0)\tilde{E}_R^{\text{b}}y.$$

Define the fiber norm

$$\|(x, y)\|_{\mathbb{B}_R^\sharp} = \|x\| + |R - R_0|\|y\|$$

after quotienting  $\ker E_R$ . The physical range is given its inherited anisotropic distribution norm, not a tautological quotient norm.

**Lemma 1.1** (Uniform birth chart). *If  $E_{R_0}^{\text{old}}$  and  $\tilde{E}_{R_0}^{\text{b}}$  have transverse closed ranges, then, after shrinking the parameter chart, the induced realization*

$$\bar{E}_R : \mathbb{B}_R^\sharp / \ker E_R \longrightarrow \text{Ran } E_R$$

*is an isomorphism with uniformly bounded norm and inverse norm.*

*Proof.* Rescale the birth coordinate by  $z = (R - R_0)y$ . In  $(x, z)$  coordinates the realization is  $E_R^{\text{old}}x + \tilde{E}_R^{\text{b}}z$ , a continuous perturbation of a bounded-below operator on the transverse sum at  $R_0$ . The bounded inverse theorem is uniform on a smaller compact chart. Pulling the norm back gives exactly  $\|x\| + |R - R_0|\|y\|$ . No inverse of an order-one ambient birth coordinate is claimed at the point where its physical current vanishes.  $\square$

**Theorem 1.2** (Continuous physical quotient field). *The old, birth, death, face, and corner charts glue to a Hausdorff continuous field  $\{\mathcal{B}_R^{\text{phys}}\}_{R \in K}$ . The billiard transfer operators and their first  $m$  parameter derivatives act on one fixed strong/weak scale of this field and satisfy uniform Lasota–Yorke estimates.*

*Proof.* On overlaps, both charts are identified through their common physical current. Lemma 1.1 makes the transition maps uniformly bounded. A death chart is the reversed birth chart. Trace jets are triangular by codimension, and each moving-boundary derivative lowers one strong order. The usual branch estimate is therefore uniform after the renormalized coordinate is used. Compactness follows from finite chart cover, cylinder truncation, and one-order loss.  $\square$

**1.2. Periodic separation and a finite family of certificates.** Let

$$c_R(v) = (S_v \kappa_R^{(1)}, S_v \kappa_R^{(2)}, S_v \tau_R, |v|) \in \mathbb{R}^4$$

for a periodic inducing word  $v$ .

**Lemma 1.3** (Periodic span algorithm). *Suppose no nonzero affine combination of  $\kappa_R^{(1)}, \kappa_R^{(2)}, \tau_R, 1$  is a coboundary on the physical quotient. Then periodic vectors affinely span  $\mathbb{R}^4$ . Enumerating admissible primitive words therefore terminates with five words  $v_0, \dots, v_4$  whose difference determinant is nonzero.*

*Proof.* If the affine span were proper, a nonzero affine functional would be constant on every periodic sum. Apply the cylinder cohomology construction: define a transfer value along a word by subtracting that constant and use equality on all periodic loops to make it path independent. This produces a continuous coboundary, contradicting the hypothesis. Linear algebra then selects five periodic vectors. Enumeration terminates because a nonzero determinant is a finite certificate.  $\square$

**Theorem 1.4** (Finite parameter-uniform certificate family). *There are finitely many five-word packets  $\mathcal{W}_1, \dots, \mathcal{W}_M$  and open parameter sets  $K_\alpha$  covering  $K$  such that*

$$|\det D_{R,\alpha}| \geq d_\alpha > 0, \quad R \in K_\alpha.$$

*The Fourier proof chooses a packet according to the parameter chart; it does not concatenate packets into one fictitious global determinant.*

*Proof.* Lemma 1.3 gives one packet at each  $R$ . Periodic collision coordinates and sums vary continuously while the word remains admissible, so its determinant stays nonzero on a neighborhood. Compactness of  $K$  gives a finite subcover. A subordinate partition in parameter space selects the corresponding estimate without altering any orbit word.  $\square$

**1.3. A separate geometric UNI theorem.** For inverse branches  $h, k$  of a common return iterate define the temporal distance

$$\Delta_{h,k}(x) = \tau_n(hx) - \tau_n(kx).$$

**Lemma 1.5** (Dispersing two-branch derivative). *On each compact finite-horizon geometry chart there exist two admissible returned branches  $h, k$ , an unstable interval  $J$ , and  $c_{\text{uni}} > 0$  such that*

$$|\partial_x \Delta_{h,k}(x)| \geq c_{\text{uni}}, \quad x \in J.$$

*Proof.* Choose two itineraries agreeing up to one collision and leaving that scatterer through two distinct strictly convex arcs before returning to the same base. For a free flight from  $r(x)$  to  $r'(x)$ ,

$$\partial_x \tau = \widehat{r' - r} \cdot (\partial_x r' - \partial_x r).$$

Subtracting the two itineraries cancels the common legs. Strict convexity and the non-tangency of the two outgoing directions make the remaining derivative nonzero at the reference orbit. Uniform curvature, finite horizon, and a compact chart shrink  $J$  so that the lower bound is uniform. Markov specification supplies common return connectors; they contribute identical terminal legs and do not change the derivative difference.  $\square$

**Theorem 1.6** (High-frequency Dolgopyat contraction). *For the twisted physical transfer operator  $\mathcal{L}_{R,\xi,b}$ , where  $\xi$  is the two-dimensional homology frequency and  $b$  the roof frequency, there are  $C, \gamma > 0$  such that*

$$\|\mathcal{L}_{R,\xi,b}^{n(b)}\|_{\text{phys}} \leq 1 - C|b|^{-\gamma}$$

for  $|b|$  large and  $\xi$  in a fixed fundamental domain. Face and corner currents satisfy the same estimate on the renormalized field.

*Proof.* Lemma 1.5 gives an interval on which  $b\Delta_{h,k}$  crosses a fixed phase interval. Split the interval into near-opposition pieces and their complement. On a near-opposition piece,

$$|zu + wv| \leq (|z| + |w|)(1 - c\alpha^2)$$

when the phase error is at most  $\alpha$  and the two amplitudes are comparable. Bounded distortion and cone invariance preserve comparability. Nonconcentration of unstable conditional measures supplies a definite mass of such pieces. The remaining branches are controlled by Lasota–Yorke contraction. Trace jets form a finite triangular extension in each chart; the depth weights and Lemma 1.1 absorb their off-diagonal terms.  $\square$

The finite-family periodic theorem is used only to exclude exact compact frequency eigenfunctions. The derivative theorem above supplies the distinct large-frequency mechanism.

**1.4. The master vector–roof local limit formula.** Let  $S_n \in \mathbb{Z}^2$  and  $T_n \in \mathbb{R}$  be the homology sum and roof sum, centered at their exact finite saddle. Write  $\Sigma_R$  for the nonsingular three-dimensional covariance and let  $\varphi_{\Sigma_R}$  be its Gaussian density.

**Theorem 1.7** (Correctly scaled vector–roof LLT). *Uniformly for lattice targets  $k_n$  in a bounded central range and bounded Borel sets  $B \subset \mathbb{R}$  with negligible boundary,*

$$\mathbb{P}_R\left(S_n = k_n, \frac{T_n}{n} \in t_n + G_n B\right) = \frac{1}{n} \int_{\sqrt{n}(t_n - \bar{\tau}_R) + \sqrt{n}G_n B} \varphi_{\Sigma_R}\left(\frac{k_n - n\bar{k}_R}{\sqrt{n}}, u\right) du + o(n^{-1})$$

*whenever  $G_n \rightarrow 0$ ; the same formula with the corresponding Gaussian interval holds for fixed  $G_n$ . In particular, if  $\sqrt{n}G_n \rightarrow 0$ , the leading order is  $G_n n^{-1/2}$ , not  $G_n n^{-3/2}$ .*

*Proof.* Fourier inversion gives a joint two-lattice/one-continuous density of order  $n^{-3/2}$  in the roof-sum variable. The average window  $T_n/n \in t_n + G_n B$  is a roof-sum interval of length  $nG_n$ . Changing variables  $u = (T_n - nt_n)/\sqrt{n}$  contributes the factor  $\sqrt{n}$ , yielding the displayed  $n^{-1}$  Gaussian integral. Compact frequencies use Theorem 1.4; large roof frequencies use Theorem 1.6; the remaining minor arcs are exponentially small by the physical spectral gap.  $\square$

**Corollary 1.8** (Three unambiguous regimes). *If  $G_n \ll n^{-1/2}$ , the coefficient is the local-density expression  $G_n n^{-1/2}$ . If  $G_n \asymp n^{-1/2}$ , it is  $n^{-1}$  times a finite Gaussian interval mass. If  $n^{-1/2} \ll G_n \leq G_0$ , it is  $n^{-1}$  times the saturated conditional Gaussian mass. One target-dependent saddle is used in all three regimes.*

*Proof.* Apply dominated convergence to the Gaussian integral in Theorem 1.7 in the three ranges of  $\sqrt{n}G_n$ .  $\square$

## SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-eight dependency ledger. Clean compilation and automated structural checks are necessary verification gates; independent external journal review remains a separate process.

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